University of Asia Pacific
Department of Computer Science and Engineering

Course Code : CSE 312

Course Title : Data Communication & Networking Lab

Lab Report : 06

Submission Date :15.09.2025

Submitted By:

Name: Nusrat Ahmmed Ekra

Student ID : 22201251

Section: E2

Objective:

The objective of this experiment is to learn and simulate of data communication line by using optisystem software. In this experiment we use transmitter component, channel component and receiver component .

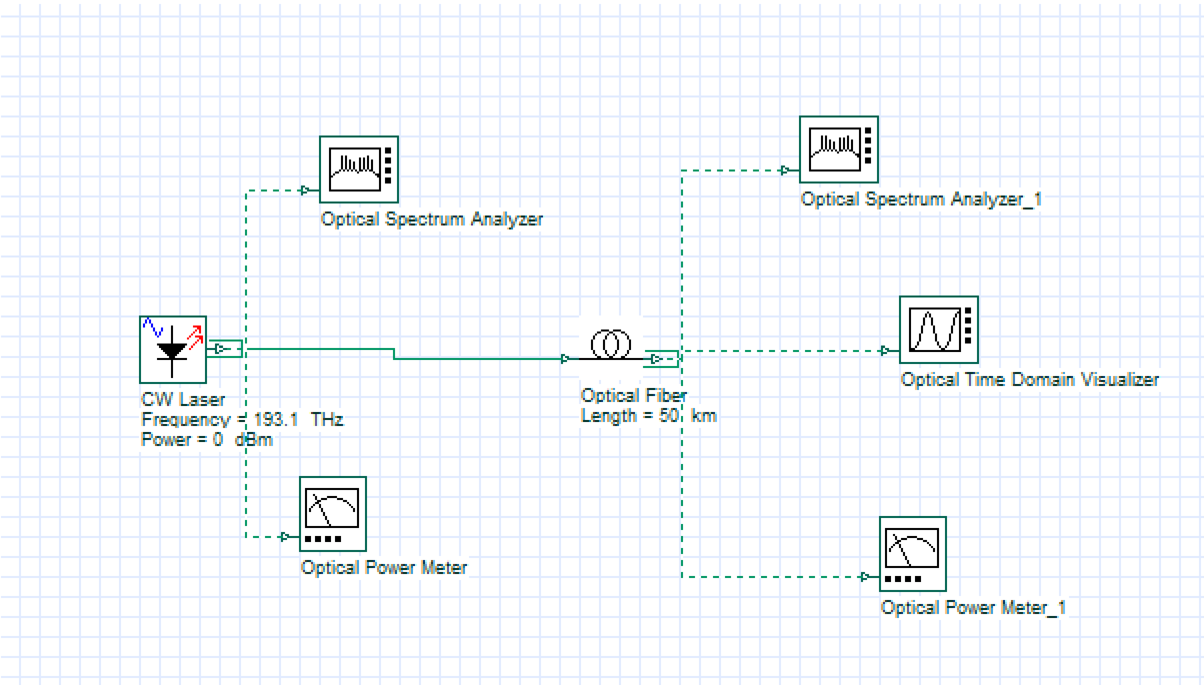
List of Components :

1. CW laser.
2. Optical fiber.
3. Optical Time Domain Visualizer.
4. Optical power meter.
5. Optical spectrum analyzer.

Theory:

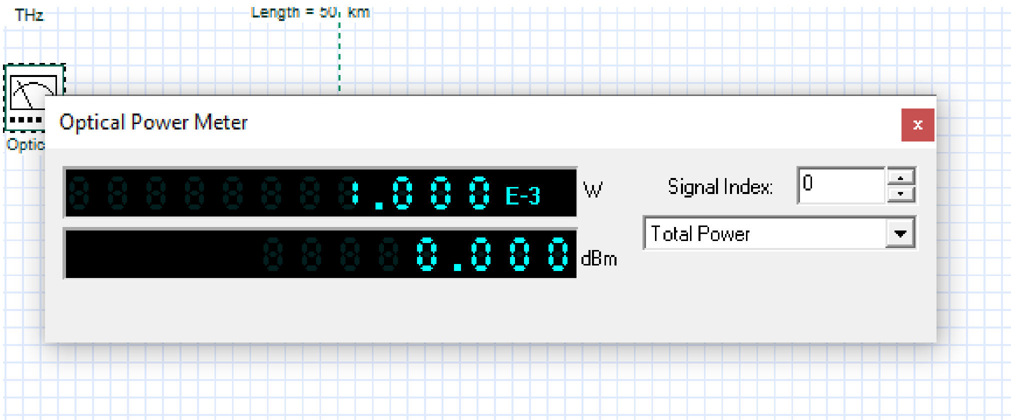
This experiment is based on introduction of optisystem that means the use of optisystem software. We create communication line through optical fiber and check the power. In this experiment we use CW laser as a transmitter component and put wavelength and power value. And then we connect a optical fiber and put also a length value. And last we receive this power value with Optical Time domain Visualizer which is a receiver component. We check power and wave length using optical power meter and spectrum meter. The purpose of this experiment is to demonstrate and investigate the wide range of function of optical communication networks.

Screenshot of the layout:

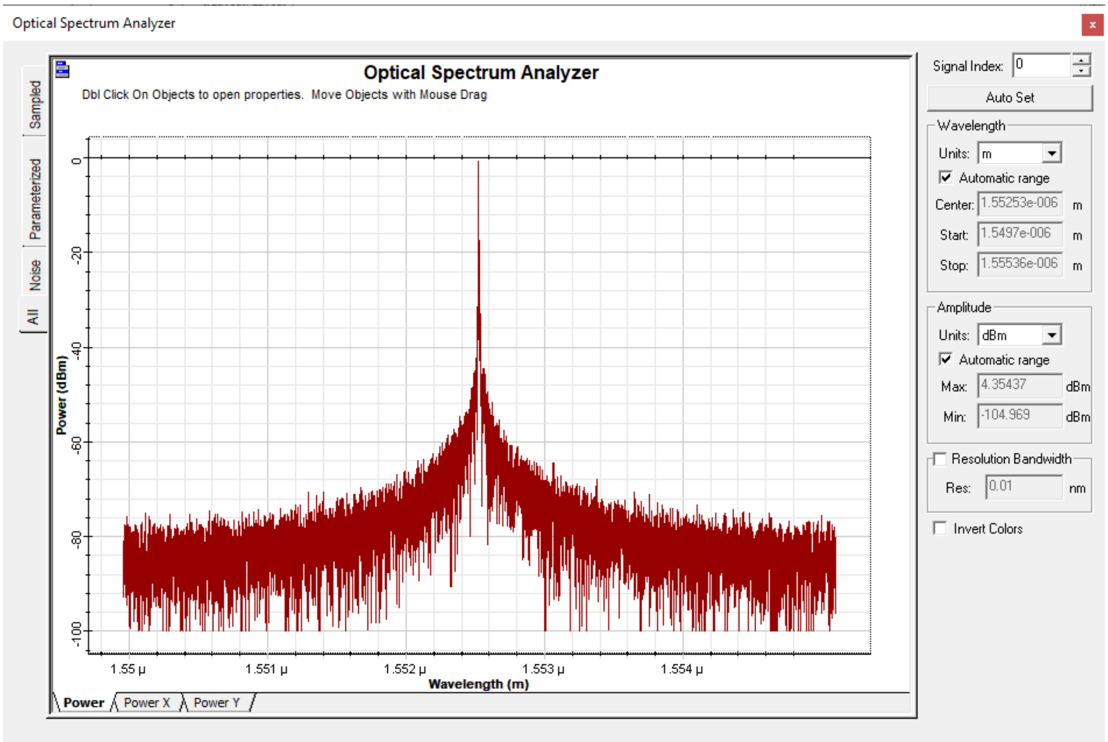


Simulation results :

Power level before optical fiber :

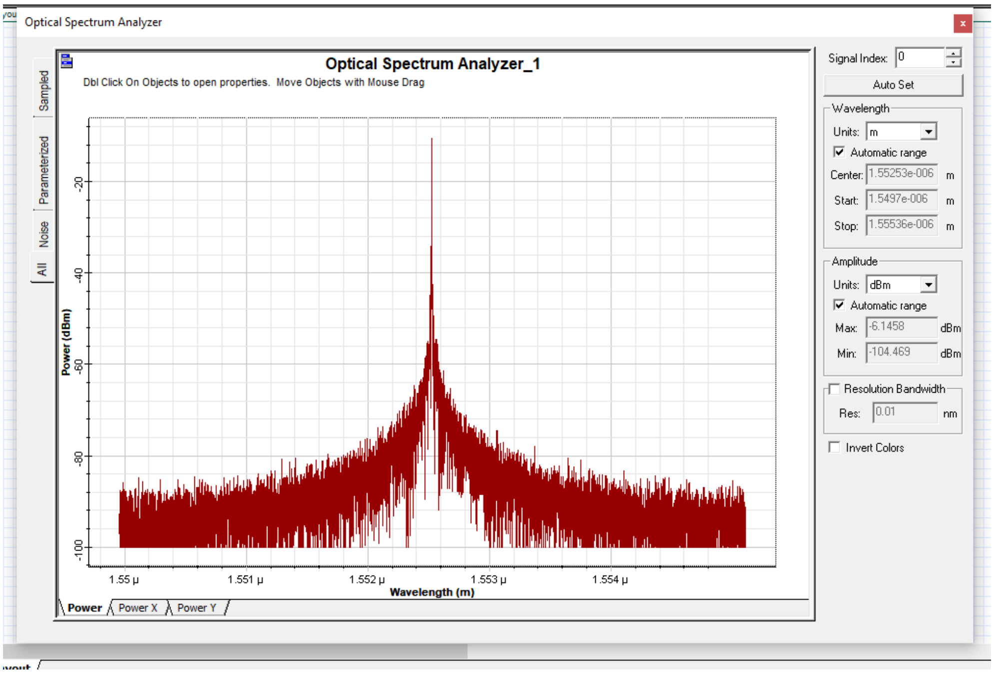


Optical Spectrum before optical fiber :



Power level After optical fiber :

Optical Spectrum After optical fiber :



Spectrum of Optical Time Domain Visualizer: